MomentNoise Documentation

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1 Introduction

MomentNoise is a noise generator whose amplitude distribution can be changed: besides the level, you set the *skewness* (asymmetry) and the *excess kurtosis* (tail weight) of the noise. The default is Gaussian (normal) noise at −24 dBFS RMS. The lower half of the window shows the theoretical probability density function (pdf) of the noise. Together with its companion plug-in AmplitudeHistogram, which measures the histogram of a signal, it shows how the statistics of a signal and its histogram belong together.

Warning: heavy-tailed noise (large excess kurtosis) has rare peaks far above its RMS level. At −24 dBFS RMS, skewness 2 and excess kurtosis 15, single peaks reach about +5 dBFS. Lower the level before you listen to such settings.

MomentNoise is a generator: the input signal is replaced by the noise (independent noise in every channel).

2 Installation

The downloaded file is a zip file. It contains the plug-in (.vst3; on macOS also the AU, .component), the Standalone application, a ReadMeFirst.txt, the licence files and this manual. Copy the plug-in to the right place and let your DAW rescan its plug-ins.

```
Windows (VST3):  C:\Program Files\Common Files\VST3
Mac (VST3):      /Users/yourUSERNAME/Library/Audio/Plug-Ins/VST3
Mac (AU):        /Users/yourUSERNAME/Library/Audio/Plug-Ins/Components
Linux (VST3):    /home/yourUSERNAME/.vst3
```

The macOS version is not signed with an Apple Developer ID. If macOS reports that the plug-in is damaged or cannot be opened, remove the quarantine flag in the Terminal, e.g. `xattr -cr ~/Library/Audio/Plug-Ins/VST3/MomentNoise.vst3`.

2.1 Presets and settings

The presets (plain XML files) and the settings file (`user.settings`: day/night theme) are stored in this folder – delete it to remove MomentNoise completely:

```
Windows: C:\Users\yourUSERNAME\AppData\Roaming\Jade\_Hochschule\MomentNoise
Mac:     /Users/yourUSERNAME/Library/Audio/Presets/Jade\_Hochschule/MomentNoise
Linux:   /home/yourUSERNAME/.config/Jade\_Hochschule/MomentNoise
```

3 How to use



1. **Preset handler**: Prev/Next step through the presets. To save a new preset, type a name and press Save. Save turns red when you change a parameter of the displayed preset; press it to update the preset. To delete a preset, delete its XML file.

2. **Day/night**: switches between the night (dark) and the day (light) theme.

3. **Skewness** $(-2 \ldots 2)$: the 3rd standardized moment. 0 is symmetric; positive values give a long tail towards positive amplitudes, negative values towards negative amplitudes.

4. **Excess kurtosis** $(-1.5 \ldots 15)$: the 4th standardized moment minus 3. 0 is Gaussian; negative values give light tails (uniform noise: -1.2), positive values heavy tails with rare large peaks (Laplace: 3). Every distribution has an excess kurtosis above skewness$^2 - 2$; MomentNoise keeps a distance of 0.5 to this bound and raises smaller values (the info text says so).

5. **Level**: RMS level (standard deviation) of the noise in dBFS.

6. **Info text**: the Johnson family used (see section 5) and the skewness and excess kurtosis that were reached.

7. **Log density**: the density axis of the display logarithmic (5 decades) or linear. The tails, where the kurtosis shows, are only visible in the logarithmic view.

8. **Display**: the theoretical pdf (red) over the standardized amplitude $(x - \text{mean})/\sigma$ from -6σ to 6σ, the normal pdf as dashed reference. AmplitudeHistogram uses the same axes, so both displays can be compared directly.

4 Controls

All controls, their range and default value. Rest the mouse on a control to see its description as a tooltip.

Control	Range	Default	Description
Skewness	-2.00 - 2.00	0.00	Asymmetry of the amplitude distribution (3rd standardized moment): 0 = symmetric, positive = long tail towards positive amplitudes.
Excess kurtosis	-1.50 - 15.00	0.00	Tail weight (4th standardized moment minus 3): 0 = Gaussian, negative = light tails (uniform: -1.2), positive = heavy tails, rare large peaks (Laplace: 3). At least skewness squared - 1.5, smaller values are raised.
Level	-60.0 dBFS - 0.0 dBFS	-24.0 dBFS	RMS level of the noise (its standard deviation). Heavy tails give peaks far above the RMS level: they may exceed 0 dBFS.
Log density	Off - On	Off	Density axis of the display logarithmic (shows the tails) or linear.

5 How it works

The noise comes from the *Johnson system* of distributions (N. L. Johnson, 1949). A standard normal random variable z is transformed into

$$x = \xi + \lambda\, y(z)$$

with one of four families:

SN (normal)	$y = z$	skewness 0, excess kurtosis 0
SL (lognormal)	$y = e^{z/\delta}$	on the "lognormal line"
SU (unbounded)	$y = \sinh(z/\delta) + t\cosh(z/\delta)$	above the lognormal line (heavy tails)
SB (bounded)	$y = 1/(e^{-z/\delta} + c)$	below the lognormal line (light tails, bounded)

($t = \tanh(\gamma/\delta)$ and $c = e^{-\gamma/\delta}$ are Johnson's parameter γ in a form that cannot overflow.) Together the families cover every possible pair of skewness and excess kurtosis, so both can be set independently. When a parameter changes, a background thread searches δ and t (or c) so that the moments of y, computed by numerical integration over z, have the wanted skewness and excess kurtosis (two nested root searches, at most about 80 ms). ξ and λ then give mean 0 and variance 1; a negative skewness mirrors the distribution. The pdf of the display follows from the change of variables, $p(x) = \varphi(z)\,|dz/dx|$.

For every sample a Gaussian random number z is drawn and transformed; the level scales the result (smoothed over 20 ms).

6 Source code

MomentNoise is open source. You find the source code here:
https://github.com/JoergBitzer/noise_moments. Please report bugs there, too.

7 Legal stuff

For this documentation the CC-BY 4.0 licence is valid. Copyright holder is Joerg Bitzer. The source code of the plug-in is under the MIT licence. The plug-in binaries contain JUCE (used under the AGPLv3), the VST3 SDK (MIT) and, on macOS, the Audio Unit SDK (Apache 2.0); the binaries as a whole are distributed under the AGPLv3.

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